

# Functional Dependencies, Minimal Cover & Normalization Proof

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## SECTION 1: ALL FUNCTIONAL DEPENDENCIES

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### 1. FARMER

FarmerID → AadhaarNo  
FarmerID → Name  
FarmerID → Phone  
FarmerID → Village  
FarmerID → Status  
AadhaarNo → FarmerID (AadhaarNo is unique per farmer)  
AadhaarNo → Name, Phone, Village, Status

### 2. COOPERATIVE\_SOCIETY

CoopID → Name  
CoopID → RegNo  
CoopID → District  
CoopID → State  
CoopID → Phone  
RegNo → CoopID (registration number is unique)  
RegNo → Name, District, State, Phone

### 3. FARMER\_MEMBERSHIP

MemberID → FarmerID  
MemberID → CoopID  
MemberID → EnrollDate  
MemberID → MemberFee  
MemberID → Status  
(FarmerID, CoopID) → MemberID (a farmer joins a coop only once)  
(FarmerID, CoopID) → EnrollDate, MemberFee, Status

### 4. CROP\_LOT

LotID → FarmerID  
LotID → CropName  
LotID → Variety  
LotID → QtyKg  
LotID → HarvestDate  
LotID → Season

### 5. PROCUREMENT

ProcureID → LotID  
ProcureID → CoopID  
ProcureID → ProcureDate  
ProcureID → QtyKg  
ProcureID → PricePerKg  
ProcureID → TotalAmt (derived: QtyKg × PricePerKg, stored for audit)

### 6. QUALITY\_CHECK

QCID → LotID  
QCID → Grade  
QCID → MoisturePct

QCID → Inspector

QCID → QCDate

## 7. FARMER\_PAYMENT

PaymentID → ProcureID

PaymentID → FarmerID

PaymentID → GrossAmt

PaymentID → Deductions

PaymentID → NetPayable (derived: GrossAmt - Deductions, stored for audit)

PaymentID → PayDate

PaymentID → Mode

## 8. WAREHOUSE

WarehouseID → CoopID

WarehouseID → Location

WarehouseID → CapacityKg

WarehouseID → Address

## 9. INVENTORY

InventoryID → WarehouseID

InventoryID → ProcureID

InventoryID → CropName

InventoryID → Grade

InventoryID → QtyAvailKg

InventoryID → ReceivedDate

InventoryID → ShelfDate

## 10. LOGISTICS\_PARTNER

PartnerID → Name

PartnerID → ServiceRegion

PartnerID → GSTIN

PartnerID → Phone

GSTIN → PartnerID (GSTIN is nationally unique)

GSTIN → Name, ServiceRegion, Phone

## 11. VEHICLE

VehicleID → PartnerID

VehicleID → VehicleType

VehicleID → CapacityKg

VehicleID → RegNo

RegNo → VehicleID (registration number is unique)

RegNo → PartnerID, VehicleType, CapacityKg

## 12. TRANSPORT\_JOB

JobID → VehicleID

JobID → ProcureID

JobID → OrderID

JobID → SchedDate

JobID → SrcAddr

JobID → AmtPayable

JobID → Status

## 13. BUYER

BuyerID → BizName  
BuyerID → Type  
BuyerID → Email  
BuyerID → GSTIN  
GSTIN → BuyerID (GSTIN is nationally unique)  
GSTIN → BizName, Type, Email

## 14. ORDER\_HEADER

OrderID → BuyerID  
OrderID → OrderDate  
OrderID → Status  
OrderID → DelivAddr  
OrderID → QtyOrdered

## 15. ORDER\_ITEM

OrderItemID → OrderID  
OrderItemID → InventoryID  
OrderItemID → QtyKg  
OrderItemID → PricePerKg  
OrderItemID → TotalAmt (derived: QtyKg × PricePerKg, stored for audit)  
(OrderID, InventoryID) → OrderItemID (same item ordered once per order)  
(OrderID, InventoryID) → QtyKg, PricePerKg, TotalAmt

## 16. CROP\_PRICE

PriceID → CoopID  
PriceID → CropName  
PriceID → Grade  
PriceID → PricePerKg  
PriceID → EffectiveDate  
PriceID → SettleAdmin  
(CoopID, CropName, Grade, EffectiveDate) → PriceID (price set once per crop/grade/date)  
(CoopID, CropName, Grade, EffectiveDate) → PricePerKg, SettleAdmin

## 17. SUBSIDY\_SCHEME

SchemeID → SchemeName  
SchemeID → Type  
SchemeID → StartDate  
SchemeID → EligibleCrop

## 18. SUBSIDY\_DISBURSEMENT

DisburseID → SchemeID  
DisburseID → FarmerID  
DisburseID → ProcureID  
DisburseID → AmtDisbursed  
DisburseID → DisDate  
DisburseID → RefNo  
DisburseID → Status

# SECTION 2: MINIMAL FD SET

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## 1. FARMER

FarmerID → AadhaarNo  
FarmerID → Name  
FarmerID → Phone  
FarmerID → Village

FarmerID → Status  
AadhaarNo → FarmerID

## 2. COOPERATIVE SOCIETY

CoopID → Name  
CoopID → RegNo  
CoopID → District  
CoopID → State  
CoopID → Phone  
RegNo → CoopID

## 3. FARMER MEMBERSHIP

MemberID → FarmerID  
MemberID → CoopID  
MemberID → EnrollDate  
MemberID → MemberFee  
MemberID → Status  
(FarmerID, CoopID) → MemberID

## 4. CROTPLOT

LotID → FarmerID  
LotID → CropName  
LotID → Variety  
LotID → QtyKg  
LotID → HarvestDate  
LotID → Season

## 5. PROCUREMENT

ProcureID → LotID  
ProcureID → CoopID  
ProcureID → ProcureDate  
ProcureID → QtyKg  
ProcureID → PricePerKg

## 6. QUALITY CHECK

QCID → LotID  
QCID → Grade  
QCID → MoisturePct  
QCID → Inspector  
QCID → QCDate

## 7. FARMER PAYMENT

PaymentID → ProcureID  
PaymentID → FarmerID  
PaymentID → GrossAmt  
PaymentID → Deductions  
PaymentID → PayDate  
PaymentID → Mode

## 8. WAREHOUSE

WarehouseID → CoopID  
WarehouseID → Location  
WarehouseID → CapacityKg  
WarehouseID → Address

## 9. INVENTORY

InventoryID → WarehouseID  
InventoryID → ProcureID  
InventoryID → CropName  
InventoryID → Grade  
InventoryID → QtyAvailKg  
InventoryID → ReceivedDate  
InventoryID → ShelfDate

## 10. LOGISTICS PARTNER

PartnerID → Name  
PartnerID → ServiceRegion  
PartnerID → GSTIN  
PartnerID → Phone  
GSTIN → PartnerID

## 11. VEHICLE

VehicleID → PartnerID  
VehicleID → VehicleType  
VehicleID → CapacityKg  
VehicleID → RegNo  
RegNo → VehicleID

## 12. TRANSPORT JOB

JobID → VehicleID  
JobID → ProcureID  
JobID → OrderID  
JobID → SchedDate  
JobID → SrcAddr  
JobID → AmtPayable  
JobID → Status

## 13. BUYER

BuyerID → BizName  
BuyerID → Type  
BuyerID → Email  
BuyerID → GSTIN  
GSTIN → BuyerID

## 14. ORDER HEADER

OrderID → BuyerID  
OrderID → OrderDate  
OrderID → Status  
OrderID → DelivAddr  
OrderID → QtyOrdered

## 15. ORDER ITEM

OrderItemID → OrderID  
OrderItemID → InventoryID  
OrderItemID → QtyKg  
OrderItemID → PricePerKg  
(OrderID, InventoryID) → OrderItemID

## 16. CROP PRICE

PriceID → CoopID  
PriceID → CropName  
PriceID → Grade  
PriceID → PricePerKg  
PriceID → EffectiveDate  
PriceID → SettleAdmin  
(CoopID, CropName, Grade, EffectiveDate) → PriceID

## 17. SUBSIDY SCHEME

SchemeID → SchemeName  
SchemeID → Type  
SchemeID → StartDate  
SchemeID → EligibleCrop

## 18. SUBSIDY DISBURSEMENT

DisburseID → SchemeID  
DisburseID → FarmerID  
DisburseID → ProcureID  
DisburseID → AmtDisbursed  
DisburseID → DisDate  
DisburseID → RefNo  
DisburseID → Status

## SECTION 3: NORMALIZATION PROOF (BCNF)

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| Relation             | Candidate Key(s)                                    | All Determinants in Minimal Cover                 | BCNF Argument                                                                                      |
|----------------------|-----------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------|
| FARMER               | {FarmerID}, {AadhaarNo}                             | FarmerID, AadhaarNo                               | Both determinants are candidate keys, hence superkeys. Every non-trivial FD has a superkey on LHS. |
| COOPERATIVE_SOCIETY  | {CoopID}, {RegNo}                                   | CoopID, RegNo                                     | Both are candidate keys. Every determinant is a superkey.                                          |
| FARMER_MEMBER_SHIP   | {MemberID}, {FarmerID, CoopID}                      | MemberID, (FarmerID, CoopID)                      | Both determinants are candidate keys of the relation. Every non-trivial FD has a superkey on LHS.  |
| CROP_LOT             | {LotID}                                             | LotID                                             | Single candidate key; LotID is the only determinant.                                               |
| PROCUREMENT          | {ProcureID}                                         | ProcureID                                         | Single candidate key. Only determinant is a superkey.                                              |
| QUALITY_CHECK        | {QCID}                                              | QCID                                              | Single candidate key. Only determinant is a superkey.                                              |
| FARMER_PAYMENT       | {PaymentID}                                         | PaymentID                                         | Single candidate key. Only determinant is a superkey.                                              |
| WAREHOUSE            | {WarehouseID}                                       | WarehouseID                                       | Single candidate key. Only determinant is a superkey.                                              |
| INVENTORY            | {InventoryID}                                       | InventoryID                                       | Single candidate key. Only determinant is a superkey.                                              |
| LOGISTICS_PARTNER    | {PartnerID}, {GSTIN}                                | PartnerID, GSTIN                                  | Both are candidate keys. Every determinant is a superkey.                                          |
| VEHICLE              | {VehicleID}, {RegNo}                                | VehicleID, RegNo                                  | Both are candidate keys. Every determinant is a superkey.                                          |
| TRANSPORT_JOB        | {JobID}                                             | JobID                                             | Single candidate key. Only determinant is a superkey.                                              |
| BUYER                | {BuyerID}, {GSTIN}                                  | BuyerID, GSTIN                                    | Both are candidate keys. Every determinant is a superkey.                                          |
| ORDER_HEADER         | {OrderID}                                           | OrderID                                           | Single candidate key. Only determinant is a superkey.                                              |
| ORDER_ITEM           | {OrderItemID}, {OrderID, InventoryID}               | OrderItemID, (OrderID, InventoryID)               | Both determinants are candidate keys. Every non-trivial FD has a superkey on LHS.                  |
| CROP_PRICE           | {PriceID}, {CoopID, CropName, Grade, EffectiveDate} | PriceID, (CoopID, CropName, Grade, EffectiveDate) | Both determinants are candidate keys. Every non-trivial FD has a superkey on LHS.                  |
| SUBSIDY_SCHEME       | {SchemeID}                                          | SchemeID                                          | Single candidate key. Only determinant is a superkey.                                              |
| SUBSIDY_DISBURSEMENT | {DisburseID}                                        | DisburseID                                        | Single candidate key. Only determinant is a superkey.                                              |

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